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SUPERSEDING
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MILITARY SPECIFICATION

VOLTAGE REGULATOR SUBASSEMBLY: 10525765

1. SCOPE

1.1 This specification covers one type of voltage regulator subassembly which, with the appropriate external circuitry, supplies -6 volt and +35 volt regulated power used in the Computer, Gun Direction: M18 for field artillery applications.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date in invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenence of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U. S. Army Munitions Command

F10525765	Voltage Regulator Subassembly

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PACKAGING DATA SHEET

M14002 Data Sheet, Master Packaging

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The voltage regulator shall be manufactured in accordance with Drawing F10525765 and drawings pertaining thereto.

3.2 General specifications.- The voltage regulator shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The voltage regulator shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the voltage regulator for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the voltage regulator shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The voltage regulator shall withstand 3 shock impulses in the vertical plane, relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (g's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The voltage regulator shall meet the requirements of 3.4.5 while exposed to temperatures of -40°F and +135°F.

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3.3.3.2 Storage.- The voltage regulator shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating.- The voltage regulator shall meet the requirements of 3.4.5 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.3.4.2 Non-operating.- The voltage regulator shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The voltage regulator shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The voltage regulator shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the voltage regulator shall meet the performance requirements of this specification at ambient temperatures between 60° and 90°F; a barometric pressure of 28 to 32 inches of mercury; and a RH up to 90%. Air at ambient temperature shall be circulated across the components at a minimum rate of 300 feet per minute.

3.4.1 Wiring.- Wiring and connections shall be in accordance with electrical schematic, Drawing F10525768. Resistors R3, R7, R27, R28, R43, R46, R64 and R69 shall be selected and assembled to the voltage regulator prior to the subsequent requirements.

3.4.2 Resistors.- After final assembly onto the voltage regulator, resistors which are not functionally part of the circuit of 3.4.5 shall possess characteristics within tolerance limits specified on the appropriate drawings.

3.4.3 Capacitors.- After final assembly onto the voltage regulator, capacitors which are not functionally part of the circuit of 3.4.5 shall possess nominal characteristics within the tolerance limits specified on the appropriate drawings.

3.4.4 Diodes.- After final assembly onto the voltage regulator, diodes which are not functionally part of the circuit of 3.4.5 shall possess characteristics within tolerance limits specified on the appropriate drawings.

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3.4.5 Circuit outputs.- The circuits in this assembly are part of two regulated power supplies. When connected to the appropriate external circuitry and when energized at input voltages corresponding to high and low AC line voltage and when supplying maximum and minimum load currents or overload currents, they shall have outputs as shown in Table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables II, III and IV and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table II shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

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TABLE II
(100 percent inspection)

	Characteristic	Requirements	Test Procedures
101.	Wiring	3.4.1	4.5.2.1
102.	Resistors	3.4.2	4.5.2.2
103.	Capacitors	3.4.3	4.5.2.3
104.	Diodes	3.4.4	4.5.2.4
105.	Circuit outputs	3.4.5	4.5.2.5

4.3 Standard sampling.

4.3.1 General.- One unit shall be selected at random by a Government representative as a standard sample from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

	Characteristic	Requirements	Test Procedures
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a standard sample from each 50 produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table IV.

TABLE IV

	Characteristic	Requirements	Test Procedures
303.	Vibration	3.3.1	4.5.1.1
304.	Shock	3.3.2	4.5.1.2
305.	Humidity - operating	3.3.4.1, 3.4.5	4.5.1.3.2, 4.5.2.5
306.	Humidity - non-operating	3.3.4.2	4.5.1.3.1
307.	Salt spray	3.3.6	4.5.1.4
308.	Operating temperature	3.3.3.1, 3.4.5	4.5.1.5, 4.5.2.6
309.	Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the represented production lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (See 4.2.3).

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4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item Number		Description
1.	Voltage measuring device	0-50 Vdc, input resistance 10 megohms or greater, accuracy of ± .5% or better
2.	Resistance measuring device	0-200 ohms accuracy of ± .5% of scale reading
3.	Current measuring device	0-10 amperes, accuracy of ± .1% full scale reading
4.	Oscilloscope	Rise time of scope plus measuring probe, 50 nano seconds per centimeter, input resistance 8 megohms and 25 pico farads or better.

4.4.1.2 In addition to the test equipment specified above, utilize the power supplies, resistors, signal sources, transistors, capacitors, potentiometers, diodes and coils designated in Tables I, V, VI and VII and Figures 1 to 6.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1.

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The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the unit rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.I.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with the conditions of 3.3.2 and Procedure II of the shock test defined in MIL-F-I3926 except that impact shocks shall not exceed 5G's. The unit shall be subsequently inspected for physical damage or defects in compliance with 3.3.2. The unit shall then be inspected as specified in 4.5.1.3 prior to determining compliance with the requirements of 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 95 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.2.5 to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The unit shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2.2 to 4.5.2.5.2.3 inclusive.

4.5.1.5 Temperature (operating and storage).- After being subjected to all environmental tests specified above, the unit shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.2.6 to determine compliance with the requirements of 3.4.3 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60°to 90°F) and the unit shall be tested in accordance with 4.5.2.2 to 4.5.2.5.2.3 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 units produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

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4.5.2 Performance.

4.5.2.1 Wiring.- To determine compliance with 3.4.1, the voltage regulator shall be visually inspected to insure that all components are physically present and conform to Drawing No. 10525768. If resistors R3, R7, R27, R28, R43, R46, R64, and R69 are missing they shall be selected and installed using the procedure indicated in 4.5.2.1.1 to 4.5.2.1.4 inclusive. If the voltage regulator conforms to Drawing No. 10525768, it shall be electrically tested as indicated in 4.5.2.2 to 4.5.2.5.2.3 inclusive.

4.5.2.1.1 Selection of resistors.- The resistors listed in Table No. V(b) shall be selected as indicated in this paragraph such that the voltage regulator shall meet the requirements of 4.5.2.5. Connect the required external circuitry to the board as indicated in Figures 1, 2 and 3. Set potentiometers RI2 and R14 described in Table No. VI at approximately the middle of the resistance range. Set potentiometer R11 for minimum resistance between terminals 19B and 22B. Set potentiometer R15 for minimum resistance between terminals 31B and 30B. Set variable resistor R7 for 8,000 ohms ± 800 ohms. Adjust the +35 volt load for a resistance of 50 ohms ± 5 ohms. Adjust the -6 volt load for a resistance of 1.2 ohms ± 0.12 ohms. Set power supply No. 3 to 16.5 volts ± 0.5 volts, Power Supply No. 5 to 22 volts ± 0.66 volts, Power Supply No. 7 to 32 volts ± 0.96 volts, Power Supply No. 9 to 52 volts ± 1.56 volts, and Power Supply No. 2 to 6 volts ± 0.18 volts. Energize the power supplies of Table No. VII in the following sequence: Power Supply No. 1, 4, 2, 5, 6, 8, 9, 7, and 3 respectively. Adjust variable resistor R7 such that the frequency of the voltage waveform measured from terminal 2A to terminal 25B is 1450 cycles per second ± 58 cycles per second.

4.5.2.1.1.1 Output voltage resistors.- Adjust the +35 volt load to 0.7 amperes ± 0.035 amperes. Adjust potentiometer R14 of Fig. No. 3 such that the output voltage across the +35 volt load is as close to 35 volts as the potentiometer resolution allows. The voltage across the 35 volt load shall be 35 volts ± 0.07 volts. Adjust the -6 volt load to 5.25 amperes ± 0.26 amperes. Adjust potentiometer R12 of Fig. No. 2 such that the output voltage across the -6 volt load is as close to -6 volts as the potentiometer resolution allows. The voltage across the -6 volt load shall be -6 volts ± 0.012 volts.

4.5.2.1.1.2 Trip circuit resistors.

4.5.2.1.1.2.1 +35 Volt trip circuit resistors.- Adjust the #35 volt load to 1.3 amperes ± 0.013 amperes. Adjust potentiometer R15 of Fig. No. 3 until the circuit trips. (The circuit is in the tripped condition when the voltage across the +35 volt load is less than 2 volts whenever the load has a resistance of less than 90 ohms). De-energize all power supplies. Set the +35 volt load for a resistance of 50 ohms ± 5 ohms. Re-energize the power supplies in the sequence of 4.5.2.1.1. Slowly increase the +35 volt load until the circuit trips. The load current when tripping occurs shall be 1.3 amperes ± 0.039 amperes. If tolerance is exceeded repeat the last step with various settings of R15 until the circuit trips within tolerance. De-energize all power supplies.

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4.5.2.1.1.2.2 -6 Volt trip circuit resistors.- Adjust the loads and power supplies in accordance with 4.5.2.1.1. Energize the power supplies in the sequence given in 4.5.2.1.1. Adjust the -6 volt load to 10.0 amperes ± 0.1 amperes. Adjust potentiometer R11 of Fig. No. 2 until the circuit trips. (The circuit is in the tripped condition when the voltage across the -6 volt load is less than 2 volts whenever the load has a resistance of less than 1.5 ohms). De-energize all power supplies. Set the -6 volt load for a resistance of 1.2 ohms ± 0.12 ohms. Re-energize the power supplies in the sequence of 4.5.2.1.1. Slowly increase the -6 volt load until the circuit trips. The load current when tripping occurs shall be 10.0 amperes ± 0.3 amperes. If tolerance is exceeded, repeat the last step with various settings of R11 until the circuit trips within tolerance.

4.5.2.1.1.3 Resistor installation.- De-energize all power supplies and install the resistors selected from the appropriate column in Table No. V(a) to the nearest resistance of the potentiometers R11, R12, R14, and R15 given in Table No. VI(b).

4.5.2.2 Resistors.- To determine compliance with 3.4.2 apply 0.25 amperes DC ± 0.0025 amperes to terminal A51 with terminal A46 as the return path for current flow. The voltage from terminal A51 to terminal A46 shall be 5.5 volts ± 0.275 volts. Apply 0.2 amperes DC ± 0.002 amperes to terminal 38A with terminal 37A as the return path for current flow. The voltage from terminal 38A to terminal 37A shall be 6.0 volts ± 0.54 volts.

4.5.2.3 Capacitors

4.5.2.3.1 DC leakage current.- To determine compliance with 3.4.3 the dc leakage current shall be measured after an electrification period sufficient to achieve a stable leakage current value but in no case to exceed 5 minutes after application of the specified dc voltage to the capacitor under test. With the test circuit connected as shown in Figure No. 4 and switch SW. 4 closed, energize power supply No. 4. The DC leakage current shall be less than 60 micro amperes.

4.5.2.3.2 AC capacitance.- Open switch SW. 4 of Figure No. 4. Vac shall be a sine wave voltage of 1 ± 0.02 volts AC rms at a frequency of 120 ± 5 cycles per second and capable of supplying 0.15 amperes AC rms. The rms AC current shall be between 0.082 ± 0.168 amperes.

4.5.2.4 Diodes

4.5.2.4.1 Forward voltage.- To determine compliance with 3.4.4, apply 0.2 amperes DC ± 0.01 amperes with less than 4 percent rms ripple to the terminals specified in Table No. VIII(a) for the appropriate test position observing the polarity stated. For each test position, the voltage across the specified terminals in Table No. VIII(a) shall be less than 1.5 volts DC.

4.5.2.4.2 Reverse current.- Connect power supply No. 8 to the terminals specified in Table No. VIII(b) for the appropriate test position observing the polarity stated. For each test position, the current through the specified terminals in Table No. VIII(b) shall be less than 2.4 micro amperes.

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4.5.2.5 Circuit outputs.- Make the following connections and measurements to determine compliance with 3.4.5.

4.5.2.5.1 +35 volt and -6 volt regulator operation.- Connect the power supplies and external circuitry according to Figures 1, 2 and 3 with the potentiometers R11, R12, R14 and R15 removed. Adjust the loads and power supplies in accordance with 4.5.2.1.1. Energize the power supplies in the sequence given in 4.5.2.1.1.

4.5.2.5.1.1 Output voltage measurements.- Output voltage measurements shall be made in accordance with tests No. 1, 2, 3, 4, and 5 of Table No. I. Remove all connections from terminals A11 and A35. Perform test No. 6 of Table No. I. Make a connection between terminals B5 and A5 and between terminals A32 and A31. Perform test No. 7 of Table No. I. Remove the connections between terminals B5 and A5 and between terminals A32 and A31. Make a connection between terminals A11 and A5 and between terminals A35 and A31.

4.5.2.5.1.2 Remote sensing resistors.- Remove all connections from terminal A1, and from terminal B54. Remove all connections between the +35 volt load and the terminals A31 and A35. Remove the connection between the -6 volt load and the terminals 5A and 11A. Perform test No. 8 of Table No. I. Replace all connections.

4.5.2.5.1.3 Trip circuit

4.5.2.5.1.3.1 +35 volt trip circuit.- Increase the +35 volt load until the circuit trips. The +35 volt load current when tripping occurs shall be 1.3 amperes ± 0.15 amperes. (At no time shall the load current exceed 1.6 amps). The voltage from terminals 12A to 39B shall be greater than 3 volts. De-energize all power supplies.

4.5.2.5.1.3.2 -6 volt trip circuit.- Adjust the loads and power supplies in accordance with 4.5.2.1.1. Energize the power supplies in the sequence given in 4.5.2.1.1. Increase the -6 volt load until the circuit trips. The -6 volt load current when tripping occurs shall be 10.0 amperes ± 0.9 amperes. (At no time shall the load current exceed 12 amperes). The voltage from terminals 8A to 25B shall be less than 2 volts. De-energize all power supplies.

4.5.2.5.1.4 Voltage transient.- Adjust the loads and power supplies in accordance with 4.5.2.1.1. Energize power supplies No. 1, 4, 2, 5, 6, 8, and 7. Energize power supply No. 3 and measure the voltage transient across the -6 volt load. The voltage shall not exceed 8 volts at any time. Transient shall be over in 3 seconds. De-energize all power supplies.

4.5.2.5.1.5 Thermistors.- Connect lead A of Figure No. 5 to terminal 57A and lead B of Figure No. 5 to terminal 65B. Energize power supply No. 1. Measure 0.77 ± 0.33 volts D.C. from terminals 57A to 65B. De-energize power supply No. 1 and connect lead A of Figure No. 5 to terminal 4B and lead B of Figure No. 5 to terminal 3A. Energize power supply No. 1. Measure 0.088 ± 0.04 volts DC from terminals 4B to 3A. De-energize power supply No. 1.

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4.5.2.5.2 Master trip circuit.- Adjust the loads and power supplies in accordance with 4.5.2.1.1. Energize the power supplies in the sequence given in 4.5.2.1.1.

4.5.2.5.2.1 Minimum bias voltage.- Adjust power supply No. 2 to 5.1 volts ± 0.051 volts. Increase the -6 volt load until the circuit trips. The voltage from terminal 8B to terminal 25B shall be less than 2 volts. De-energize the power supplies.

4.5.2.5.2.2 Maximum bias voltage.- Adjust the loads and energize the power supplies in accordance with 4.5.2.1.1. Adjust power supply No. 2 to 6.9 volts ± 0.069 volts. Increase the +35 volt load until the circuit trips. Open the switch SW. 1 of Fig. No. 1. The voltage from terminal 58A to terminal 60A shall be less than 5 volts. De-energize the power supplies.

4.5.2.5.2.3 Transient voltage.- Remove the connections from terminals 56B, 41A, and 56A. Open switch SW. 2. Energize power supply No. 5. Close switch SW. 2. The voltage from terminal 58A to terminal 60A shall be greater than 5 volts. De-energize power supply No. 5. Make a connection from terminal A62 to terminal A63. Open switch SW. 1. Remove the connections from termfnal 12A. Remove all connections from 8B and 37B. Energize power supply No. 5. Close switch SW. 3. The voltage from terminal 58A to terminal 60A shall be greater than 5 volts.

4.5.2.6 Extreme environmental conditions.- Utilize the setup specified in 4.5.2.5.1 to conduct the following tests.

4.5.2.6.1 Low operating temperature - (40°F).- Performance at the low operating temperature specified in 3.3.3.1 shall be determined as follows to assure compliance with 3.4.5.

4.5.2.6.1.1 Output voltage measurement.- Output voltage measurements shall be made in accordance with tests No. 9, 10, 11 and 12 of Table No. I.

4.5.2.6.1.2 Trip circuit.

4.5.2.6.1.2.1 +35 volt trip circuit.- Perform the tests of 4.5.2.5.1.3.1 except that the +35 volt load current when tripping occurs shall be 1.3 amperes ± 0.26 amperes.

4.5.2.6.1.2.2 -6 volt trip circuit.- Perform the tests of 4.5.2.5.1.3.2 except that the -6 volt load current when tripping occurs shall be 10.5 amperes ± 1.47 amperes.

4.5.2.6.1.3 Master trip circuit.- Perform the instructions of 4.5.2.5.1. Perform the tests of 4.5.2.5.2.1, 4.5.2.5.2.2 and 4.5.2.5.2.3.

4.5.2.6.1.4 Capacitors.- Perform the test of 4.5.2.3.2 except that the rms AC current shall be between 0.053 and 0.168 amperes.

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4.5.2.6.2 High operating temperature (+135°F).- Performance at the high operating temperature specified in 3.3.3.1 shall be determined in accordance with 4.5.2.6 above and the following, to determine compliance with 3.4.5.

4.5.2.6.2.1 Output voltage measurement.- Output voltage measurements shall be made in accordance with tests No. 9, 10, 11 and 12 of Table No. I.

4.5.2.6.2.2 Trip circuit.

4.5.2.6.2.2.1 +35 volt trip circuit.- Perform the tests of 4.5.2.5.1.3.1 except that the +35 volt load current when tripping occurs shall be 1.3 amperes ± 0.26 amperes.

4.5.2.6.2.2.2 -6 volt trip circuit.- Perform the tests of 4.5.2.5.1.3.2 except that the -6 volt load current when tripping occurs shall be 10.0 amperes ± 1.47 amperes.

4.5.2.6.2.3 Master trip circuit.- Perform the instructions of 4.5.2.5.1. Perform the tests of paragraphs 4.5.2.5.2.1, 4.5.2.5.2.2 and 4.5.2.5.2.3.

4.5.2.6.2.4 Diodes.- Perform the test of 4.5.2.4.2 except that the current through the specified terminals in table No. VIII shall be less than 60 micro amperes.

4.5.2.6.2.5 Capacitors.- Perform the test of 4.5.2.3.2 except that the rms AC current shall be between 0.082 and 0.218 amperes.

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TABLE I

Output Voltages and Load Currents

Test Number	Input dc Power Supply Voltage Settings (± 2 percent)				Load Current Settings		Output Voltage Measurements		Maximum Output *Ripple Measurements	
	Power Supply No. 3	Power Supply No. 5	Power Supply No. 7	Power Supply No. 9	+35v load ± 5% (Amperes)	-6v load ± 5% (Amperes)	+35v load (Volts dc)	-6v load (Volts dc)	+35v load (Volts)	-6v load (Volts)
1	16.5	22	32	52	0.7	5.25	35 ± 0.35	-6 ± 0.06	0.05	0.3
2	13	18.0	25	44	0.39	3.8	35 ± 0.7	-6 ± 0.18	0.05	0.3
3	13	18.0	25	44	1.02	6.7	35 ± 0.7	-6 ± 0.18	0.05	0.3
4	20	26	37.5	62.5	0.39	3.8	35 ± 0.7	-6 ± 0.18	0.05	0.3
5	20	26	37.5	62.5	1.02	6.7	35 ± 0.7	-6 ± 0.18	0.05	0.3
6	16.5	22	32	52	0.7	5.25	36.75±0.75	-6.3±0.13	----	---
7	16.5	22	32	52	0.7	5.25	33.25±0.75	-5.7±0.13	----	---
8	16.5	22	32	52	0.7	5.25	35 ± 1.0	-6 ± 0.6	----	---
9	13	18	25	44	0.39	3.8	35 ± 1.1	-6 ± 0.3	0.05	0.3
10	13	18	25	44	1.02	6.7	35 ± 1.1	-6 ± 0.3	0.05	0.3
11	20	26	37.5	62.5	0.39	3.8	35 ± 1.1	-6 ± 0.3	0.05	0.3
12	20	26	37.5	62.5	1.02	6.7	35 ± 1.1	-6 ± 0.3	0.05	0.3

*NOTE: Ripple is defined as peak-to-peak (p-p) value of any alternating component of the output voltage excluding noise of less than 10 microseconds duration.

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TABLE V (a)

RESISTOR SELECTION LIST [1/]

Nominal Value (Ohms)	Part No.
5	10526702
6	10526703
6.7	10526704
8.4	10526705
10	10526706
11.7	10526707
13.4	10526708
15	10526709
16.7	10526710
18.4	10526711
20	10526712
21.7	10526713
23.4	10526714
25	10526715
26.7	10526716
28.4	10526717
30	10526718
35	10526719
40	10526720
45	10526721
50	10526722
55	10526723
60	10526724
65	10526725
70	10526726
75	10526727
80	10526728
85	10526729
90	10526730
95	10526731
100	10526732
105	10526733

[1/] R3, R7, R27 and R28 vary between 5 and 105 ohms.
R43, R46 and R64 vary between 5 and 65 ohms.

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TABLE V (b)

Resistor No.	Potentiometer arm resistance connected from Terminal	to Terminal
R3	B19	B22
R7	B23	B19
R27	B38	B31
R28	B30	B31
R43	A34	B50
R46	B52	B50
R64	B6	B10
R69	B6	B11

NOTE: Resistor R3 plus resistor R7 must equal 110 ohms ± 0.5 percent and resistor R27 plus resistor R28 must equal 110 ohms ± 0.5 percent.

Resistor R43 plus resistor R46 must equal 70 ohms ± 0.5 percent and resistor R64 plus resistor R69 must equal 70 ohms ± 0.5 percent.

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TABLE VI

EXTERNAL CIRCUITRY COMPONENT VALUES

(a) Ordnance Numbered Parts

COMPONENT NUMBER	ORDNANCE NUMBER	COMPONENT NUMBER	ORDNANCE NUMBER
* Q1, Q2, Q5, Q6, Q7, Q8	10525749	R6	10526080
** Q3, Q4, Q9	10525764		
Q5	10525843	R8	8622834-31
Q6	10525796	R9	10526631
*** CR1	10525738	R10	10526670
CR2, CR3	10525850		
CR4	10525848		
CR5, CR6, CR7, CR8, CR9, CR10, CR11	10525737		
C1, C7, C8, C9, C10	10525817	R13	10525799
C2, C3	10525754	R16	10525778-17
C4	10525820	R17	10525778-18
C5	10525865		
C6	10525786		
L1	10525930		
R1	10525778-2		
R2, R3	10525875		
R4	10525778-5		
R5	10525778-7		

(b) Potentiometers and Resistors

COMPONENT NUMBER	COMPONENT TYPE	COMPONENT VALUE (OHMS)	TOLERANCE (PERCENT)	WATTAGE (WATTS)
R7	Variable Resistor	5K to 11K	0.5	0.5
R11	Potentiometer	110	0.5	0.5
R12	Potentiometer	70	0.5	0.5
R14	Potentiometer	70	0.5	0.5
R15	Potentiometer	110	0.5	0.5
R18	Fixed Resistor	82.5 K	1.0	1/8

* These transistors shall be mounted on heat dissipator Ordnance No. 10525739.

** These transistors shall be mounted on the transistor mounting bracket Ordnance No. 10525736.

*** This diode shall be mounted on the diode mounting bracket Ordnance No. 10525776.

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TABLE VII

D. C. POWER SUPPLIES

POWER SUPPLY NO.	NOMINAL VOLTAGE (VOLTS)	VOLTAGE TOLERANCE (PERCENT)	MAXIMUM RMS RIPPLE (M VOLTS)	CURRENT REQUIREMENTS (AMPERES)	MAXIMUM OUTPUT IMPEDANCE (OHMS)	TERMINAL CONNECTIONS POSITIVE-TERM.	NEGATIVE-TERM.
1	6	± 5	10	0.2	1	25B	55A
2	4 to 8 *	± 2	10	0.2	1	SW 1	25B
3	10 to 30 *	± 2	500	15	0.5	See Fig.No. 2	---
4	12	± 5	10	0.2	1	See Fig.No. 2	---
5	15 to 30 *	± 2	500	1.5	1	29B, SW2, SW3	25B
6	27	± 5	500	0.2	1	25B	26B
7	20 to 40	± 2	500	2.5	0.5	See Fig.No. 2	---
8	35	± 2	20	0.1	1	See Fig.No. 2	---
9	(See Fig.No. 6)						

* Power supplies shall be continuously variable throughout the voltage specified.

NOTE: All power supplies shall be electrically isolated from each other except as designated under terminal connections. Switches are shown on Figure No. 1.

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TABLE VIII

(a) Diode Forward Voltage

Test Position	Voltage Regulator Positive Terminals	Voltage Regulator Negative Terminals
1	22A, 23A	24A, 25A
2	17B, 18B	24A, 25A
3	19A, 20A	24A, 25A
4	13B, 14B	22A, 23A
5	13B, 14B	17B, 18B
6	13B, 14B	19A, 20A
7	48A	43A

(b) Diode Reverse Voltage

Test Position	Voltage Regulator Positive Terminals	Voltage Regulator Negative Terminals
8	24A, 25A	22A, 23A
9	24A, 25A	17B, 18B
10	24A, 25A	19A, 20A
11	22A, 23A	13B, 14B
12	17B, 18B	13B, 14B
13	19A, 20A	13B, 14B
14	43A	48A

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.

6.1.1 +35 volt series regulator.- Part of this assembly consists of the sensing and amplifier portions of a +35 volt series regulators. With the output power transistors connected to the voltage regulator the +35 volt series regulator is designed to meet the requirements of this specification for a load of 0.39 to 1.02 amperes.

6.1.2 -6 volt pulse width modulated regulator.- Part of this assembly consists of the sensing and amplifier portions of a -6 volt pulse width modulated regulator. With the output power transistors and the filtering portions connected to the voltage regulator the -6 volt pulse width modulated regulator is designed to meet the requirements of this specification for a load of 3.8 to 6.7 amperes.

6.1.3 Master trip circuit.- Part of this assembly consists of a master trip circuit. With the master trip circuit connected to the +155, +35, +15, +6, -6, -10, and -18 volt regulators, the master trip circuit will turn off all of the above regulators whenever an over current is sensed by any one of the above regulators. The master trip circuit remains tripped until it is reset by removing the power from the master trip circuit.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Selection of applicable level of preservation, packaging and packing

(c) Applicable stock number

(d) Applicable packaging data sheet (see 5.1).

(e) A block of serial numbers to identify each voltage regulator.

Custodian:

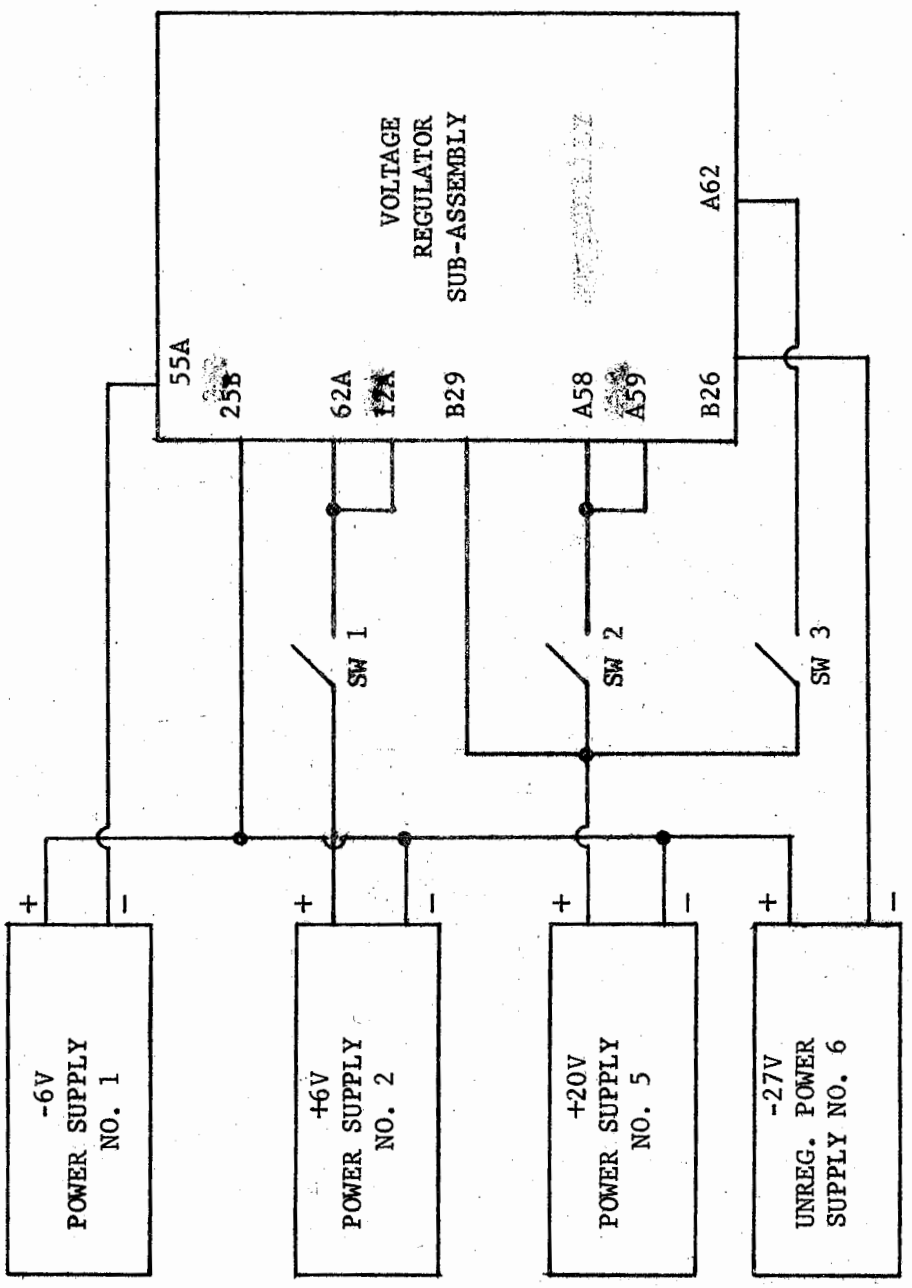
Army - MU

Preparing Activity:

Army - MU

Project No. 1220-A197Y

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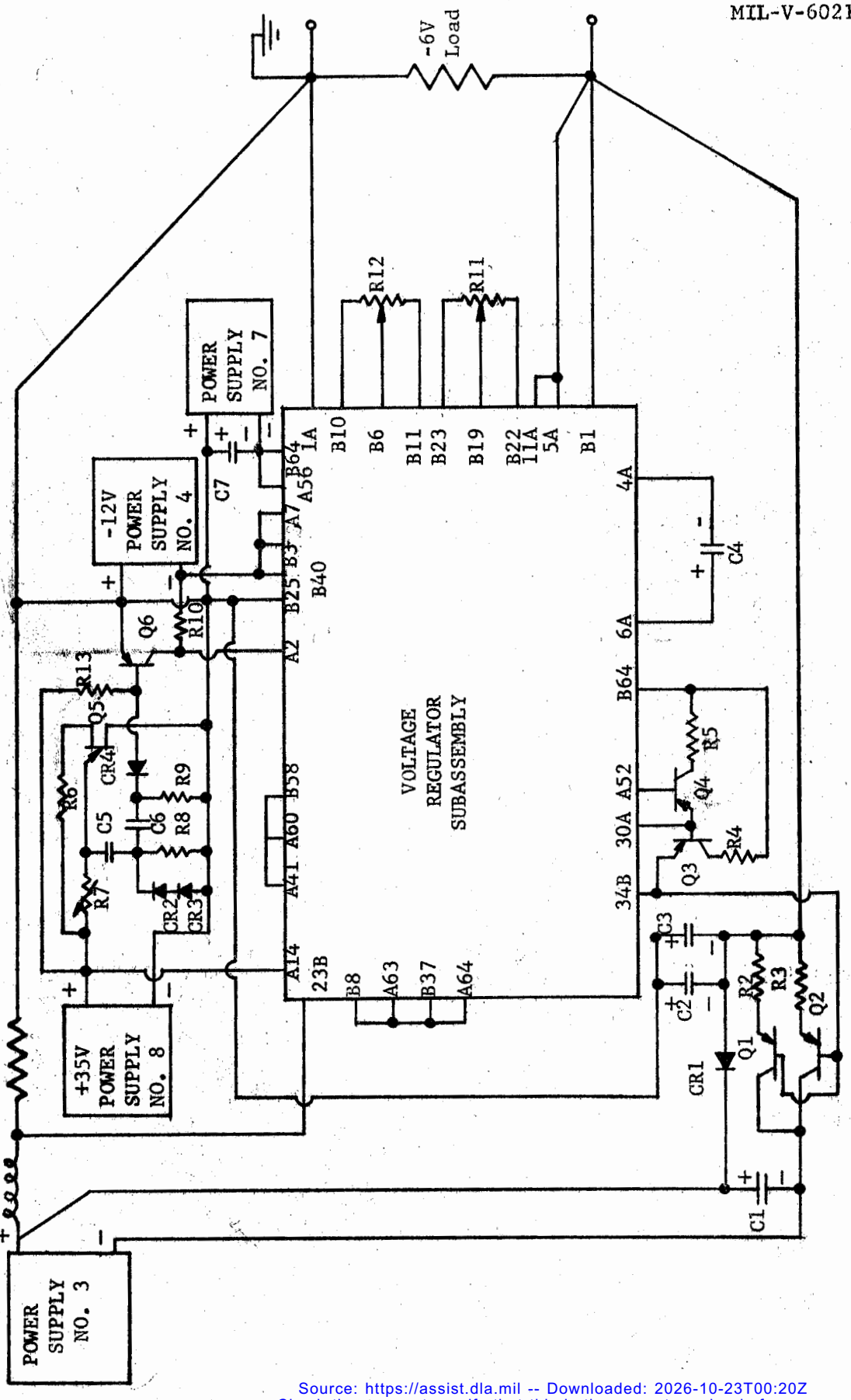


NOTE: Unless otherwise specified, switches SW1 and SW2 shall be closed and switch SW3 shall be open.

FIGURE NO. 1

POWER SUPPLY CONNECTIONS

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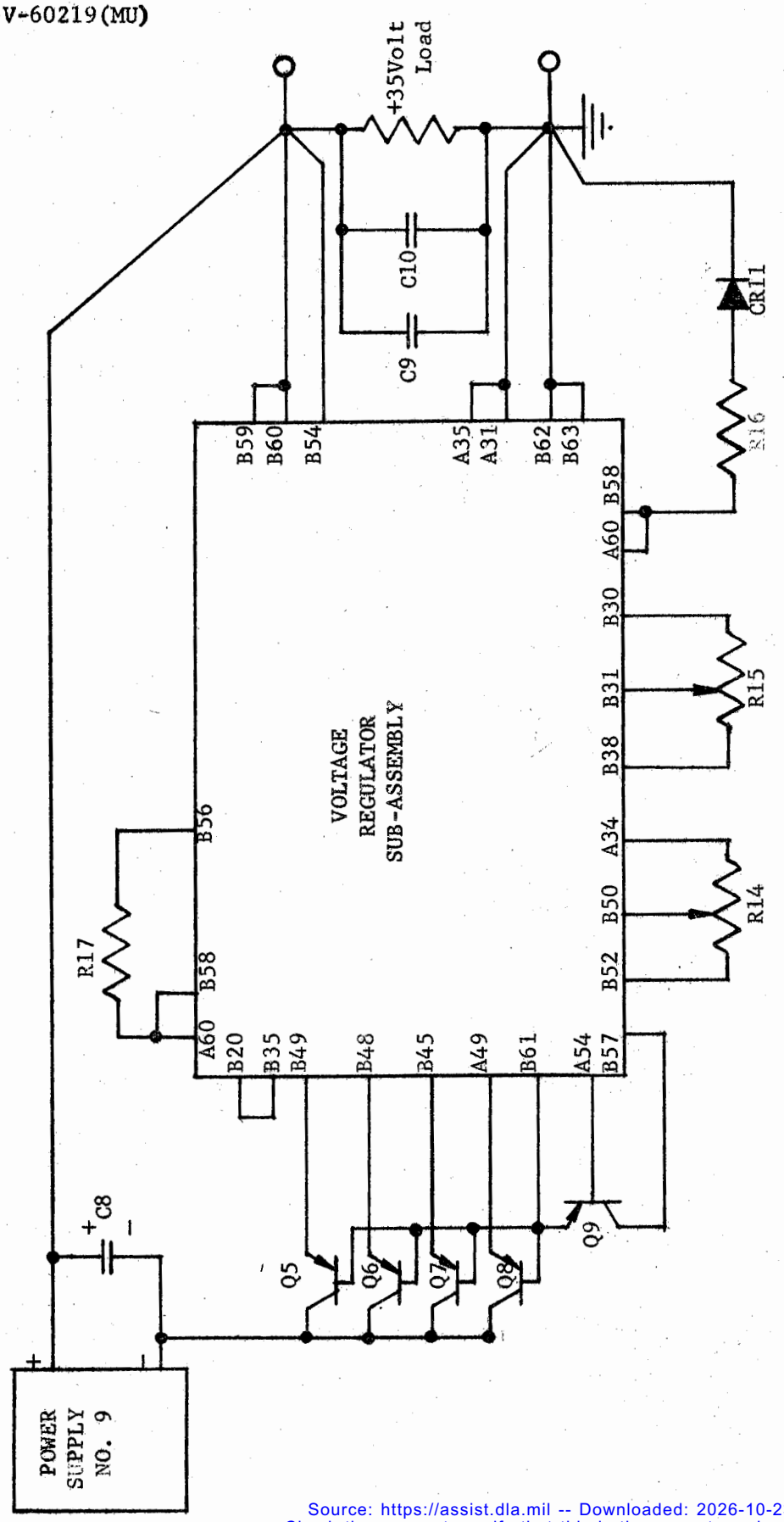


NOTE: See Table No. VI and Table No. VII for part descriptions

FIGURE NO. 2

-6 Volt Regulator Connections

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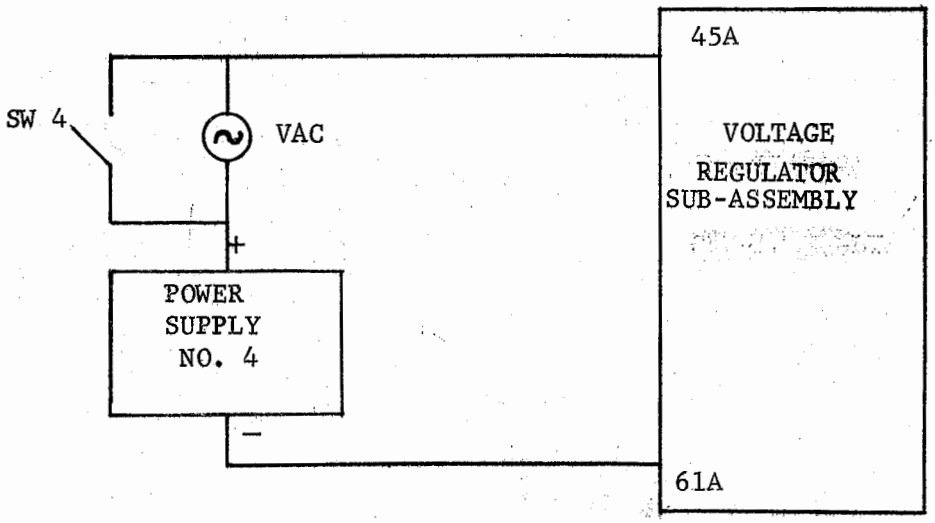


NOTE: See Table No. VI and Table No. VII for part descriptions.

FIGURE NO. 3

+35 Volt Regulator Connections

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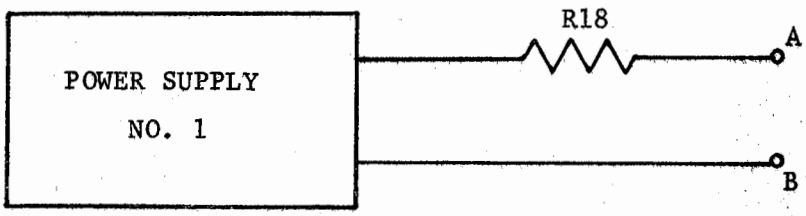
NOTE: The circuit shown in Figure 4 is given only as a guide. Any test circuit configuration that achieves the test conditions called for in paragraphs 4.5.2.3.1 and 4.5.2.3.2 may be used.

Vac shall not be energized when switch SW. 4 is closed.

FIGURE NO. 4

CAPACITOR TEST CIRCUIT

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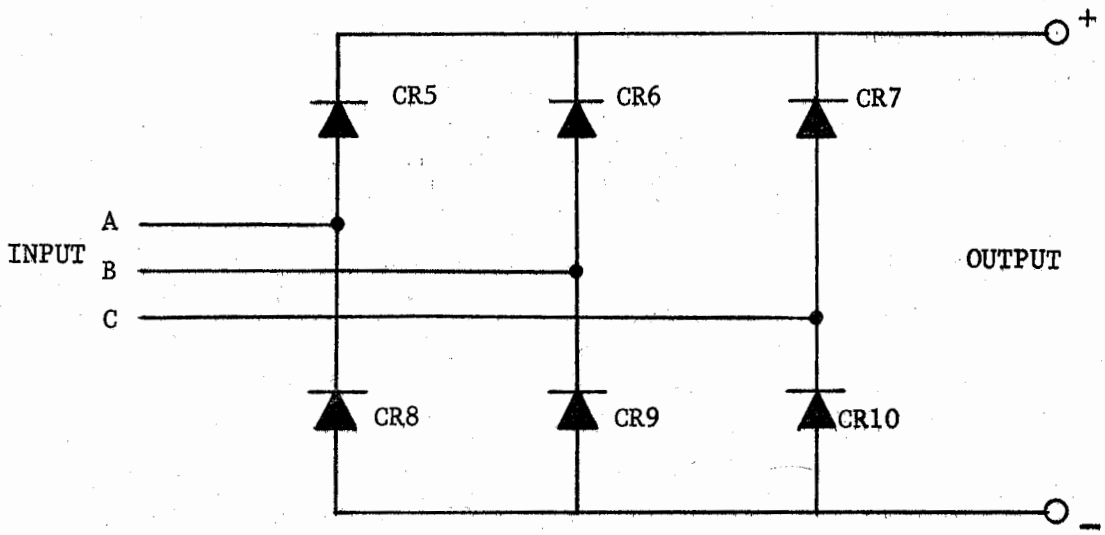


NOTE: See Table No. VI and Table No. VII for part descriptions.

FIGURE NO. 5

CONNECTIONS FOR CHECKING THERMISTERS

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INPUT:	3 phase:	AC Voltage
	Frequency:	400 ± 20 cycles per second
	Line Balance:	The voltage variation between any two lines shall not exceed 8 percent of the average of the three line to line voltages.
OUTPUT:	Voltage:	40 to 65 volts DC continuously variable.
	Current:	1.7 amperes DC maximum

The peak to peak (p-p) ripple* of the output shall not exceed 20 percent of the average direct current output voltage.

*Ripple is herein defined as peak-to-peak (p-p) value of any alternating component of the Output Voltage excluding noise of less than 10 microsecond duration.

FIGURE NO. 6

POWER SUPPLY NO. 9

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